MATH129: Big Data
HW 6
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Problem 1: Murphy 11.2 - EM for mixtures of Gaussians

- maximum likelihood estimation using EM algorithm

1. expected complete-data log-likelihood

- Gaussian Mixture Model: $p(x_i) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)$
- in EM algorithm, M-step maximizes: $Q(\theta) = \sum_i \sum_k r_{ik} [\log \pi_k + \log \mathcal{N}(x_i | \mu_k, \Sigma_k)]$
- For a particular component k, ignoring terms constant w/ respect to μ_k and Σ_k:

$$Q_k = -\frac{1}{2} \sum_i r_{ik} [\log |\Sigma_k| + (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k)].$$

2. Derive mean update

$$\frac{\partial Q_k}{\partial \mu_k} = \sum_i r_{ik} [\Sigma_k^{-1} (x_i - \mu_k)]$$

$$\left(0 = \sum_i r_{ik} [\Sigma_k^{-1} (x_i - \mu_k)] \right) \Sigma_k$$

$$0 = \sum_i r_{ik} (x_i - \mu_k)$$

$$0 = \sum_i r_{ik} x_i - \mu_k \sum_i r_{ik}, \text{ since } r_k = \sum_i r_{ik}$$

$$\boxed{\mu_k = \frac{\sum_i r_{ik} x_i}{r_k}}$$

3. Derive covariance update

Let $A_k = \sum_i r_{ik} (x_i - \mu_k)(x_i - \mu_k)^T$ then covariance-dependent objective becomes:

$$Q_k = -\frac{1}{2} [r_k \log |\Sigma_k| + tr(\Sigma_k^{-1} A_k)].$$

Define $\Lambda_k = \Sigma_k^{-1}$. Since $\log |\Sigma_k| = -\log |\Lambda_k|$, $Q_k = \frac{r_k}{2} \log |\Lambda_k| - \frac{1}{2} tr(A_k \Lambda_k)$.

$$\frac{\partial Q_k}{\partial \Lambda_k} = \frac{r_k}{2} \Lambda_k^{-1} - \frac{1}{2} A_k$$

$$0 = \frac{r_k}{2} \Lambda_k^{-1} - \frac{1}{2} A_k$$

$$\left(\frac{1}{2} A_k = \frac{r_k}{2} \Lambda_k^{-1} \right) 2$$

$\rightarrow A_k = r_k \Lambda_k^{-1}$. Since $\Lambda_k^{-1} = \Sigma_k$,

$$\boxed{\sum_k = \frac{1}{r_k} \sum r_{ik} (x_i - \mu_k)(x_i - \mu_k)^T}$$

4. Derive alternate covariance expression

expanding outer product:

$$(x_i - \mu_k)(x_i - \mu_k)^T = x_i x_i^T - x_i \mu_k^T - \mu_k x_i^T + \mu_k \mu_k^T$$

$$\therefore \Sigma_k = \frac{1}{r_k}\left[\sum_i r_{ik} x_i x_i^T - \left(\sum_i r_{ik} x_i\right)\mu_k^T - \mu_k\left(\sum_i r_{ik} x_i\right)^T + r_k \mu_k \mu_k^T\right].$$

recall $\sum_i r_{ik} x_i = r_k \mu_k$, $\Sigma_k = \frac{1}{r_k}\left[\sum_i r_{ik} x_i x_i^T - r_k \mu_k \mu_k^T\right]$

$$\boxed{\Sigma_k = \frac{1}{r_k}\left[\sum_i r_{ik} x_i x_i^T - \mu_k \mu_k^T\right]}$$ □

Problem 2: SVD Image Compression

for image matrix $A \in \mathbb{R}^{m \times n}$, the SVD is $A = U \Sigma V^T$.

the k-rank approximation is, $A_k = U_{[:,1:k]} \Sigma_{[1:k,1:k]} V^T_{[1:k,:]}$.

Code implementation attached.